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Blockchain-Driven Queueing and Transparent Budget Allocation for Scholarships

 **CC 103 - Data Structures and Algorithms**
CS 1206

 JRC Studios

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🌟 Overview

ScholarChain is a console-based C++ application that simulates a **scholarship allocation system** using queueing and blockchain-inspired data structures.

It demonstrates the practical use of **data structures and algorithms**, including stacks, queues, and priority queues, alongside recursion and iteration for efficient processing.

Users can:

-  Apply for scholarships
-  View ranked applicants
-  Process applicants fairly
-  Manage waitlisted entries

Data Handling

-  Applicant records are processed and stored using structured data systems.
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🌟 Features

1. **Applicant Evaluation.** Assesses eligibility based on academic and financial criteria
 2. **Priority Ranking.** Uses priority queue for fair distribution
 3. **Queue System.** Processes applicants in FIFO order
 4. **Blockchain Record.** Stores approved applicants chronologically
 5. **Budget Allocation.** Allocates funds until limit is reached
 6. **Waitlist System.** Handles overflow applicants
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🌟 How to Run the Program

Open your terminal in the project folder and run:

```
g++ main.cpp queue.cpp priority.cpp stack.cpp hashing.cpp network.cpp -o dsa
```

Run the program using:

```
./dsa
```

🌟 Project Structure

```
DSA/
├── main.cpp
├── README.md
├── .gitignore
├── Data Structure/
│   ├── queue.cpp
│   ├── priority.cpp
│   └── stack.cpp
└── Integrity/
    ├── hashing.cpp
    └── network.cpp
```

- **main.cpp** - Entry point of the program; handles user interaction and system flow
- **.gitignore** - Specifies files to be ignored by Git

📁 Data Structure

- **queue.cpp** - Handles normal applicant processing (FIFO)
- **priority.cpp** - Manages high-priority applicants
- **stack.cpp** - Stores blockchain-like records

📁 Integrity

- **hashing.cpp** - Ensures data integrity using hashing techniques
- **network.cpp** - Simulates blockchain network validation/consensus

🌟 Example Output

```
Welcome Admin!
```

```
Please set budget before processing!
```

```
Budget: 100000
```

```
ScholarChain Menu:
```

```
1: Process Scholarship
2: Push Queue to Blockchain
3: Show Transactions
4: Show Queue
5: Show Remaining Budget
6: Sort Transaction
7: Verify Transaction Integrity
8: Add Budget
9: Exit Program
Choice (1 - 9): 1
```

```
Processing Scholarships
```

```
Enter Student ID Number: 25-09705
```

```
Enter Student Name: Jobine Arago
```

```
Enter Student GPA: 2.50
```

```
Enter Student/Family Income: 50000
```

```
Enter Scholarship Grant: 2000
```

```
Applicant is queued for the normal queue.
```

```
Would you like to process another? (Y/N):
```

☞★.°Contributors

Name	Role
Charles V. Amelano	Main Programmer
Mary Jobine R. Arago	Documenter
Ralph Ivan I. Bituin	System Architect

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